

Chapter 1: Introduction to Operating Systems

An operating system (OS) is system software that manages computer hardware, software resources, and provides a common interface for users to interact with the computer system.

Key Functions:

- Process Management
- Memory Management
- File System Management
- I/O Management

Chapter 2: Process Management

A process is a program in execution. It includes the program code, current activity, and allocated resources.

Process States:

1. New - The process is being created.
2. Ready - The process is waiting to be assigned to a processor.
3. Running - Instructions are being executed.
4. Waiting - The process is waiting for some event.
5. Terminated - The process has finished execution.

Chapter 3: Deadlocks

A deadlock is a situation where a set of processes are blocked because each process is holding a resource

Banker's Algorithm:

The Banker's Algorithm is a resource allocation and deadlock avoidance algorithm that tests for safety by simulating the allocation of